Team 18

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Team 18

Content Overview

01 Intro Slide

02 Content Overview

03 Challenge Area Addressed

04 Data Overview

05 Analytical Approach

06 Key Finding 1

07 Key Finding 2

08 Implications For Burn Care

09 Recommendations

10 Future Work/Closing Remarks

Use Case

Challenge Area 3: Advancing Equitable Access to Burn Care

Clinical Challenge

- Where do structural inequities limit access to timely burn care?

Key Questions

- Where is designated burn care capacity currently lacking?
- Who is impacted by this lack of access?
- Who is most in need of these services?

Data Overview

Data Sources and Variables:

1. NIRD Dataset
   a. Total Beds
   b. Trauma Adult & Trauma Peds
   c. Burn Beds & ABA-verified & State-Designated Burn Center
   d. Location & Address
2. U.S. Census Bureau (2020)
   a. Population Estimates
   b. Population Percent Change
3. USDA Forest Service (2024)
   a. Wildfire Risk

Analytical Approach

Methods Used:

- Excel

  Burn bed distribution and total bed distribution by state and region of the United States

  Bed distributions with

- R

  Comparison of various health facilities between Maine and Montana

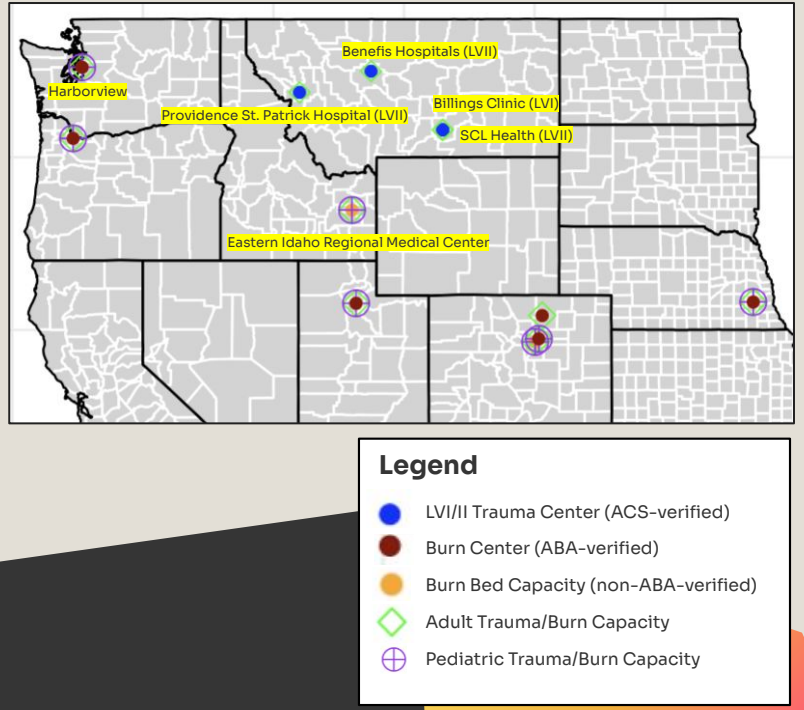
  Spatial distributions of trauma centers and burn centers in the United States

Key Finding #1
Northwest Rural States Lack Burn Care Capacity

1. Regional burn capacity in the northwestern US, especially in Montana, North Dakota, South Dakota, and Wyoming, is nonexistent.

2. Burn care centers in the Northwest are overburdened from serving multiple states:
   a. Eastern Idaho Regional Medical Center serves ID, MT, ND, & WY (not ABA certified)
      i. 318 hospital beds → 15 burn beds
   b. Harborview in Seattle, WA is a regional trauma and burn referral center for AK, ID, MT, & WA (ABA-certified)
      i. 413 hospital beds → 40 burn beds

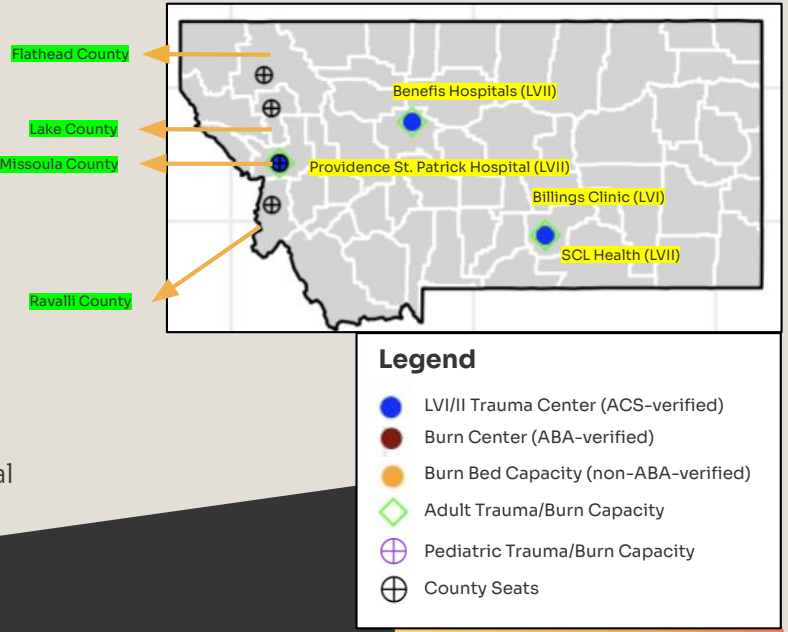
Trauma Centers (LVI & LVII) and Burn Centers in Northwestern United States



Key Finding #2
Western Montana Counties Have High Need for Burn Care

1. Montana's four trauma centers lack pediatric trauma capacity.[1]
2. Montana's population density is concentrated in its western and southern regions, but the western region lacks a Level I trauma center.[2]
3. Flathead, Lake, Ravalli, and Missoula Counties have high risk for wildfires and densely populated, rapidly growing cities[3]
   a. For each county:
      i. Wildfire risk > 89-99% of U.S. counties[3]
      ii. Wildfire risk > 75-100% of Montana counties[3]
   b. Ravalli County and Flathead County
      i. Population Percent Change: 9.1-9.7%[4]
   c. Lake and Missoula County
      i. Population Percent Change: 0.6-3.9%[4]
4. Travel distances from these counties[5]
   a. Range from 47 to 121 miles to Missoula
   b. Range from 366 to 440 miles to Eastern Idaho Regional Medical Center
   c. Range from 486 to 521 miles to Harborview

Trauma Centers (LVI & LVII) and County Seats in Montana



Implications for Burn Care

Why This Matters:

- We identified a region of the US (the northwestern states of Montana, North Dakota, South Dakota, and Wyoming) that is underserved because of no designated burn centers or dedicated burn beds in the region.
- We discovered a relatively high density population area in western Montana that is distant and isolated from the state's only Level I trauma center and identified four counties to prioritize for improved burn care, based on the relatively significant size of the population that lacks burn resources and high risk from wildfires.
- Lack of pediatric trauma designation worsens outcomes for children (source)
- We found similarities between Maine and Montana that make burn care innovation in Maine potentially relevant as a recommendation for Montana.
    a. State population size is similar for Maine and Montana.
    b. Both states must refer to out-of-state burn centers for the most severe or high-risk burn cases and stand to gain from more regionalized care

Recommendations

Potential Solutions

1. Invest funding towards Patrick St. Providence Hospital (LVII Trauma Center) in Missoula, to build designated burn capacity and a stronger regional system in Western Montana with increased capacity to manage mild to moderate burn injuries within the region.

2. Reduce burden and referral rate to more distant, out-of-state burn centers like Harborview and Eastern Idaho Medical Center (these centers are even further for Montana residents and are also serving multiple states).
   - See example from Maine, a state with a population of similar size to Montana's, demonstrating that stronger regional care systems improve burn outcomes (doi.org/10.1097/BCR.0b013e3182343dab)

3. Invest in pediatric trauma capacity in Patrick St. Providence Hospital to improve evaluation and treatment outcomes for children in Western Montana
   - There is a predicted minimum of 25% of victims from any mass casualty incident will be children (WHCC Burn Surge Annex)
   - We probably need to specify the percentage of children in our counties of interest

Future Work

Next Steps

1. Analyze best available data on best practices and outcomes for regional triage, management and referral for burn cases, including in Maine.
2. Compare regional burn case management practices and outcomes before and after implementation of the recommended regional triage, management and referral solution in western Montana.
3. Identify other regions in the US with similar need and characteristics, as additional candidates for the recommended regional solution.

Closing Remarks

Thank you to the American Burn Association and BData for organizing the Heat Map Hackathon.

Thank you to the University of Minnesota for hosting the final ceremony.

Purpose: Evaluates how well the team framed and addressed a real-world problem within their selected use case.

Primary Use Case Identification & Alignment (1–5)

Clearly identifies one primary use case
Aligns problem statement, data, and solution

Use Case–Driven Insights (1–5)

Analysis generates insights relevant to the selected use case
Findings are supported by evidence
Actionability is demonstrated

Patient, Provider, or System Impact (1–5)

Demonstrates relevance to real-world stakeholders
Impact may be clinical, operational, or systemic

Purpose: Evaluates technical rigor, appropriateness, and creativity of the approach.

Methodological Soundness (1–5)

Methods are appropriate, correct, and well explained
Assumptions are reasonable and justified

Innovation & Creativity (1–5)

Applies novel approaches or creative problem-solving
Goes beyond a purely standard or expected analysis

Data Integration & Insight Generation (1–5)

Effectively uses available data
Integrates context to generate defensible insights tied to the use case

Purpose: Evaluates technical rigor, appropriateness, and creativity of the approach.

Clarity & Storytelling (1–5)

Clear narrative from problem → analysis → solution
Logical and easy to follow

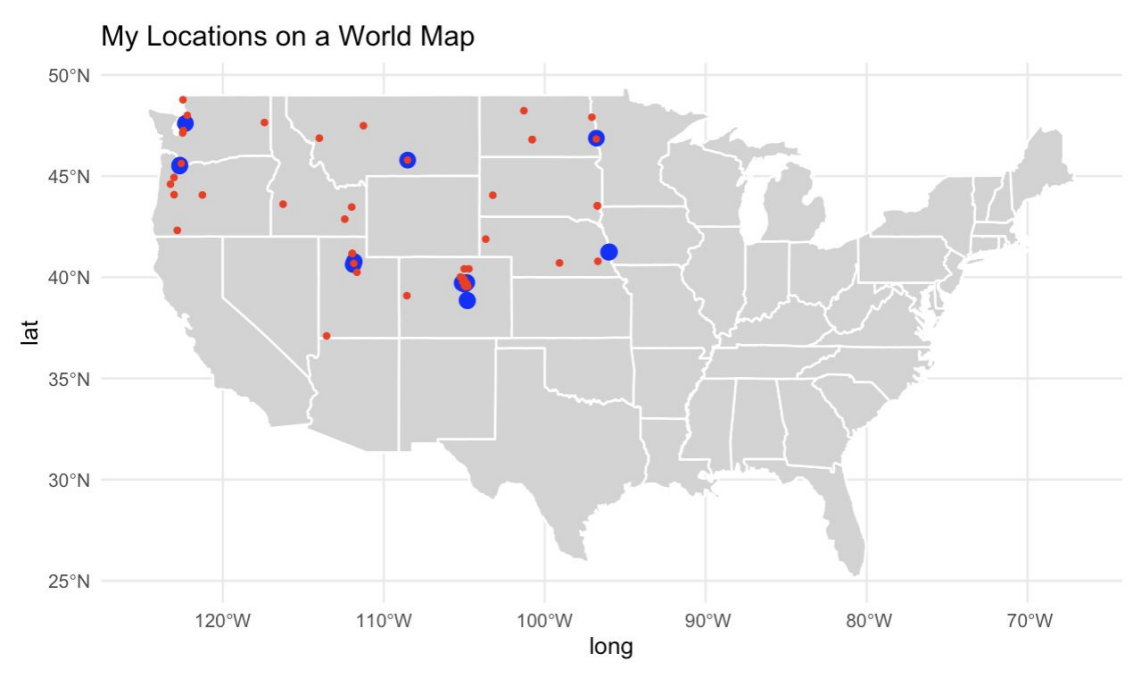
Visual & Verbal Quality (1–5)

Slides and visuals are clear and professional
Verbal explanation is understandable and engaging
Competition rules and required components were followed

Actionability, Feasibility & Impact (1–5)

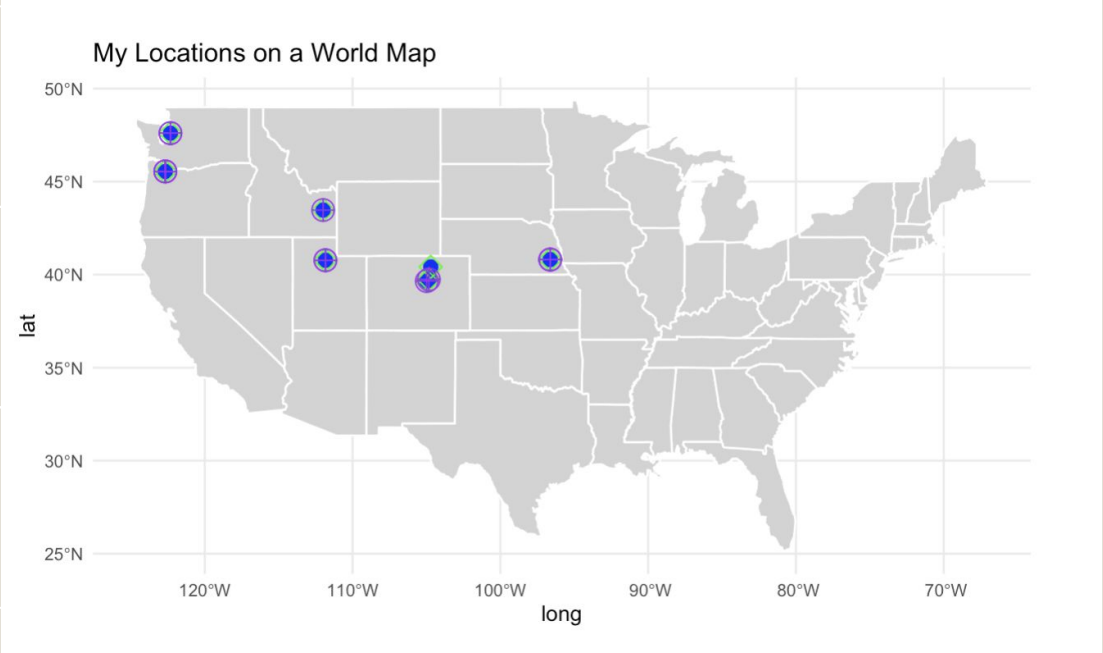
Clear, realistic next steps
Feasible implementation considerations
Potential for real-world impact

Trauma Centers in NW United States



All LV I and LV II trauma centers in WA, OR, ID, UT, MO, WY, NE, CO, ND, SD
Blue = LV I trauma centers
Red = LV II trauma centers

Trauma Centers in NW United States



All centers with burn beds/ burn capacity in WA, OR, ID, UT, MO, WY, NE, CO, ND, SD
Green = adult burn bed capacity
Purple = ped burn bed capacity
Summary: all centers with burn beds have pediatric capacity except for one in colorado

LINKS:

- **Injury Mortality In Montana, 2014-2023:**
  - https://dphhs.mt.gov/assets/publichealth/EMSTS/Data/Injury_Mortality_2014_2023.pdf
- **Montana Injury Prevention Annual Report**
  - https://archive.legmt.gov/content/Committees/Interim/2023-2024/Transition-Review-Commission/Meetings/March-2024/2-Montana_Injury_Prevention_Annual_Report_2020.pdf
- **Montana Wildfire History Archive**
  - https://archive.legmt.gov/lfd/wildfire/
- **Montana Trauma Report 2022: A summary of 2021 data**
  - https://dphhs.mt.gov/assets/publichealth/EMSTS/Data/TraumaReport_2021Data.pdf
- **Geographic Access to Burn Center Hospitals**
  - https://pmc.ncbi.nlm.nih.gov/articles/PMC3045670/

Basis for slide 03

Use Case

Challenge Question: Where do structural inequities limit access to timely burn care?

Key Questions

- Where is designated burn care capacity currently lacking?
- Who is impacted by this lack of access?
- Who is most in need of these services?

Basis for slide 04

Data Sources

Data Sources:

1. NIRD dataset
2. U.S. Census Bureau
3. USDA Forest Service

Variables Used in Analysis

Examples:

-

Basis for slide 05

Methods

Analytical Approach- Methods Used

Describe the analytical methods used.

Examples:

- Exploratory data analysis
- Geographic mapping
- Network analysis of referrals
- Regression modeling
- Clustering or segmentation

Tools

Examples:

- Python / R
- Tableau / Power BI
- GIS mapping tools
-

Basis for slide 06

Key Finding #1: Rural States in the Northwest Lack Burn Care Capacity

BONIFACE: comparison of total beds distribution between regions (west vs south vs north vs east)

Key Finding #1: Northwest Region of the United States

Example of Title:

NW Region of U.S; especially Montana

Uneven Geographic Distribution of Burn Care

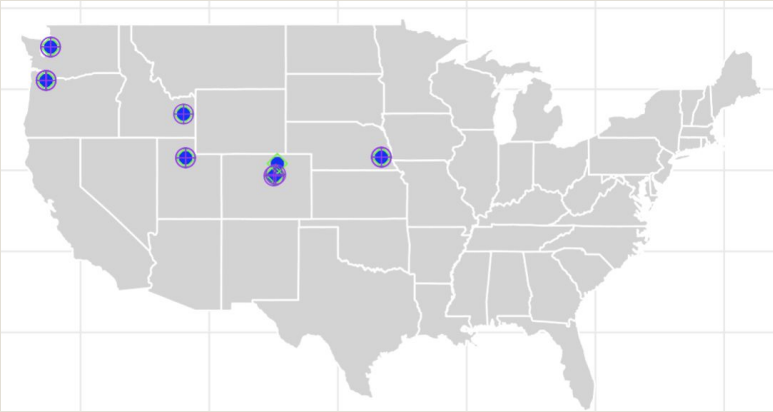
Finding

Example:

- Certain regions show significantly fewer burn center referrals relative to patient volume.

Evidence

- Map visualization
- Summary statistics



Include **one key figure or visualization**

Basis for slide 07

Key Finding #2

Key Finding #2: Insight Title

Example of Title:

Referral Patterns Between Hospitals

Finding

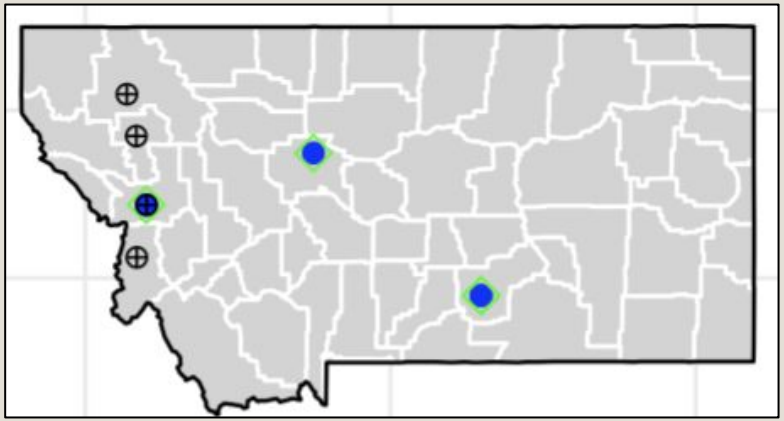
Example:

- Most burn patients are transferred within regional networks.
- A small number of hospitals serve as referral hubs.

Evidence

- Network visualization
- Transfer frequency statistics

Include **one key visualization**



Basis for slide 08

Implications for Burn Care

Why This Matters: Explain how your findings could improve burn care.

Examples:

- Identify underserved regions
- Improve referral guidelines
- Inform telemedicine deployment
- Guide resource allocation

Focus on **practical clinical impact**